

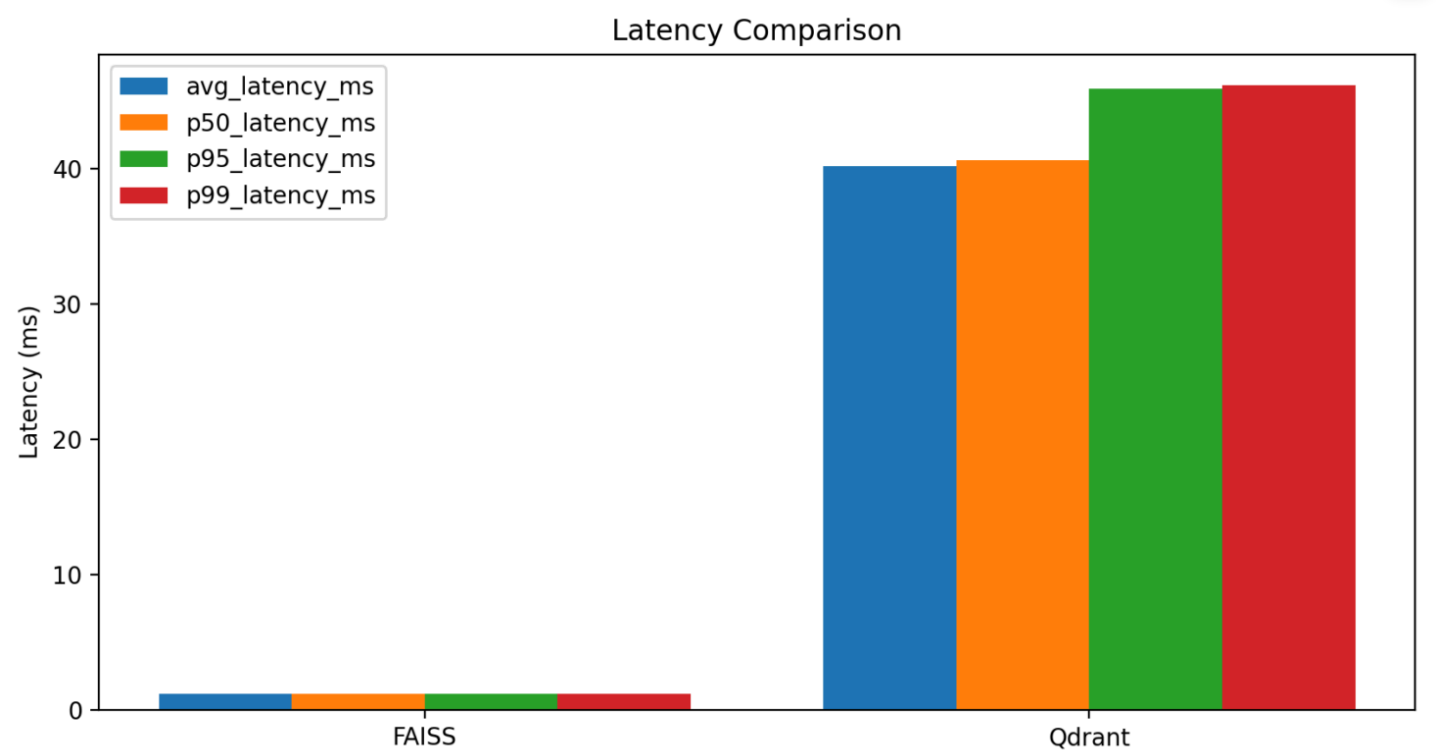
VectorDB Arena

Semantic Similarity Search Benchmark

Benchmark Summary

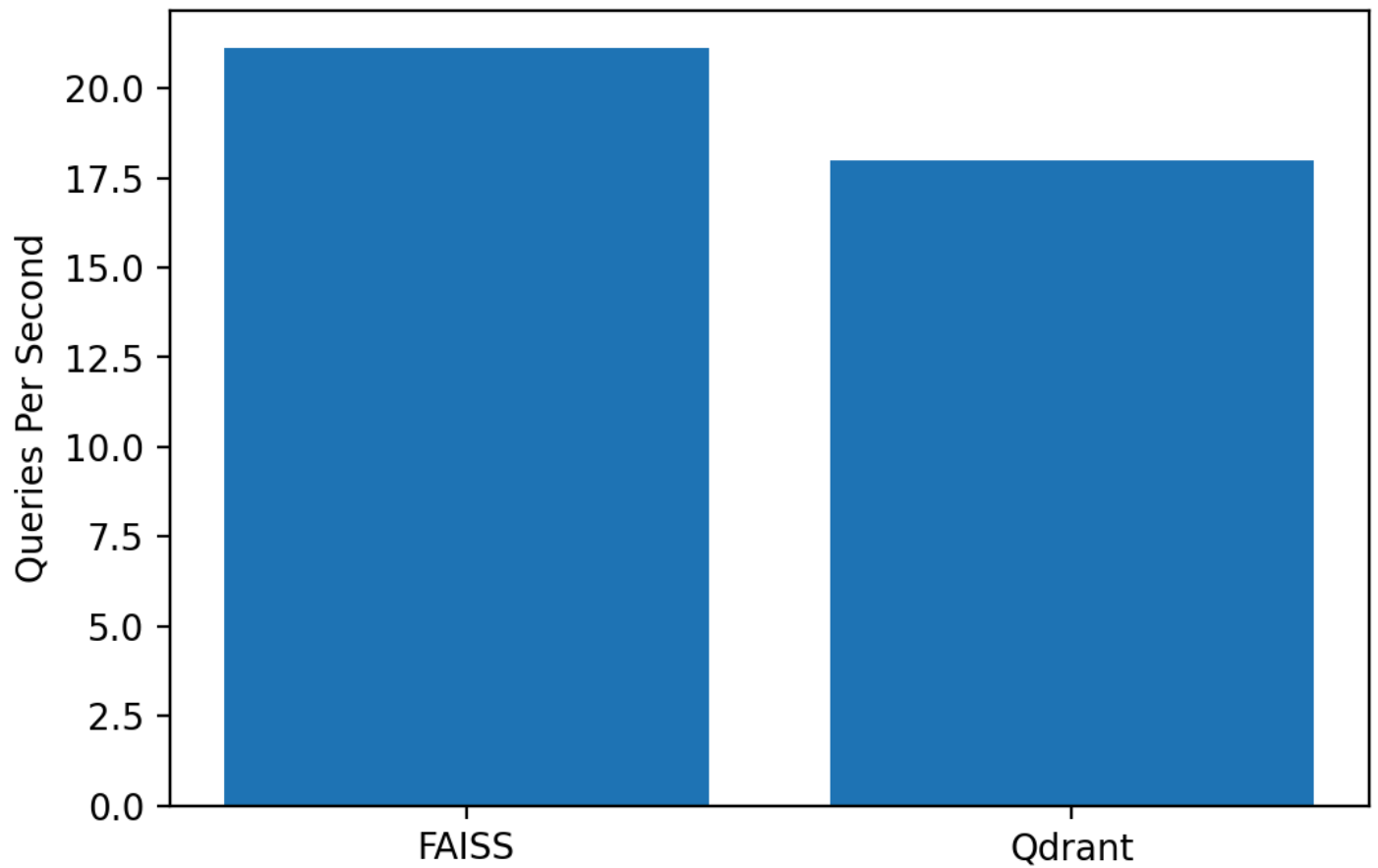
	system	dataset_size	top_k	num_queries	index_build_time_sec	avg_latency_ms	p50_latency_ms	p95_latency_ms	p99_latency_ms
0	FAISS	10000	5	4	0.0208	1.1791	1.1806	1.187	1.187
1	Qdrant	10000	5	4	3.9368	40.1584	40.6256	45.9377	46.148

Latency Metrics



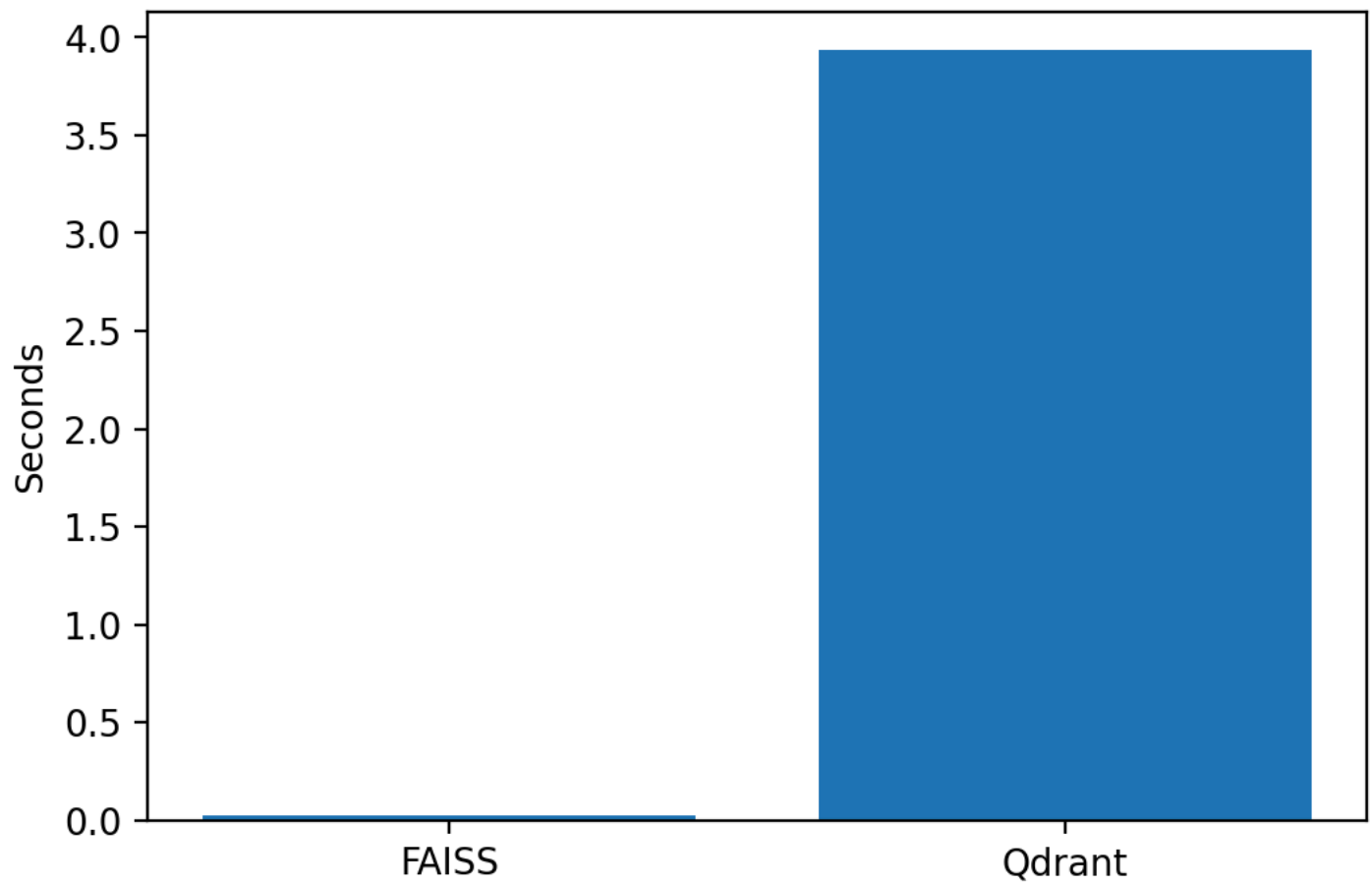
Throughput (QPS)

Throughput Comparison



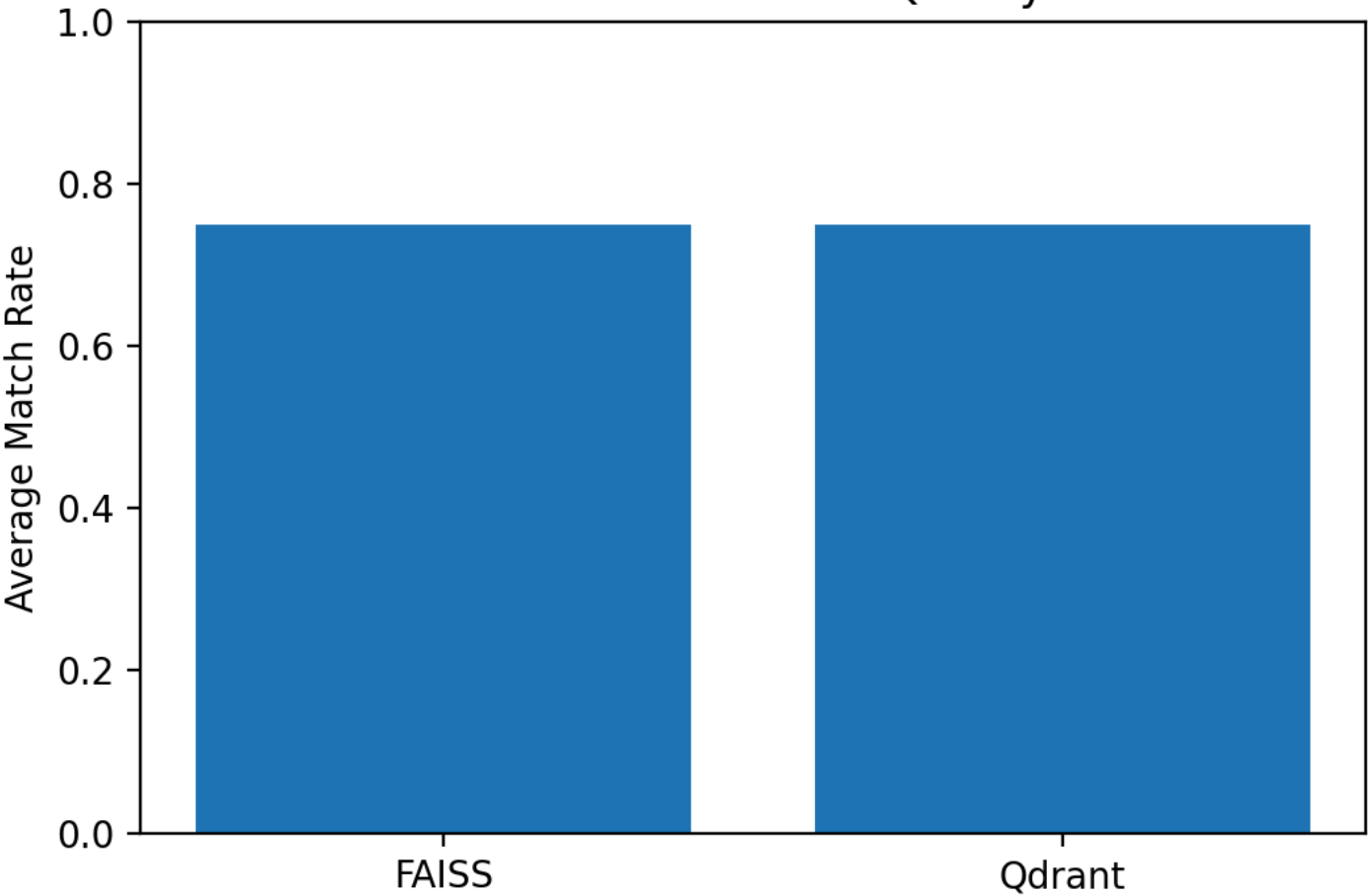
Index Build Time

Index Build Time



Category Match Rate

Semantic Retrieval Quality



Detailed Query Results

	system	query	expected_category	latency_ms	top_k	category_match_rate	top_result
0	FAISS	latest technology and artificial intelligence	3	1.1678	5	0.8	NASA Develops Robust
1	FAISS	global politics and international news	0	1.1875	5	0.6	Online Satirists Pull No
2	FAISS	stock market and business news	2	1.1845	5	0.8	Stocks Up on Conflictir
3	FAISS	football basketball and sports teams	1	1.1768	5	0.8	US no-show: Teamworl
4	Qdrant	latest technology and artificial intelligence	3	36.8088	5	0.8	NASA Develops Robust
5	Qdrant	global politics and international news	0	46.2015	5	0.6	Online Satirists Pull No
6	Qdrant	stock market and business news	2	44.4425	5	0.8	Stocks Up on Conflictir
7	Qdrant	football basketball and sports teams	1	33.1808	5	0.8	US no-show: Teamworl

Per Query Analysis

FAISS

Query: latest technology and artificial intelligence

Latency: 1.1678 ms

Category Match Rate: 0.80

Top Result: NASA Develops Robust Artificial Intelligence for Planetary Rovers NASA is planning to add a strong dose of artificial intelligence (AI) to planetary rovers to make them much more self-reliant, capable of making basic decisions during a mission. Scientists are developing very complex AI software that enables a higher level of robotic intelligence.

FAISS

Query: global politics and international news

Latency: 1.1875 ms

Category Match Rate: 0.60

Top Result: Online Satirists Pull No Punches on U.S. Election (Reuters) Reuters - Some are born to rule, others to\ridicule, is a worthy reminder to U.S presidential hopefuls\struck by an endless arsenal of satirical barbs on the Web.

FAISS

Query: stock market and business news

Latency: 1.1845 ms

Category Match Rate: 0.80

Top Result: Stocks Up on Conflicting Economic Reports NEW YORK - Stocks inched higher in quiet trading Wednesday as a pair of government reports offered conflicting signals about the economy and oil prices declined. Despite the modest rise in share prices, investors have been in no hurry to commit new money to stocks...

FAISS

Query: football basketball and sports teams

Latency: 1.1768 ms

Category Match Rate: 0.80

Top Result: US no-show: Teamwork When the NBA began sending its players to the Olympics in 1992, one of the league #39;s goals was to spread basketball around the globe. Clearly, the idea worked.

Qdrant

Query: latest technology and artificial intelligence

Latency: 36.8088 ms

Category Match Rate: 0.80

Top Result: NASA Develops Robust Artificial Intelligence for Planetary Rovers NASA is planning to add a strong dose of artificial intelligence (AI) to planetary rovers to make them much more self-reliant, capable of making basic decisions during a mission. Scientists are developing very complex AI software that enables a higher level of robotic intelligence.

Qdrant

Query: global politics and international news

Latency: 46.2015 ms

Category Match Rate: 0.60

Top Result: Online Satirists Pull No Punches on U.S. Election (Reuters) Reuters - Some are born to rule, others to\ridicule, is a worthy reminder to U.S presidential hopefuls\struck by an endless arsenal of satirical barbs on the Web.

Qdrant

Query: stock market and business news

Latency: 44.4425 ms

Category Match Rate: 0.80

Top Result: Stocks Up on Conflicting Economic Reports NEW YORK - Stocks inched higher in quiet trading Wednesday as a pair of government reports offered conflicting signals about the economy and oil prices declined. Despite the modest rise in share prices, investors have been in no hurry to commit new money to stocks...

Qdrant

Query: football basketball and sports teams

Latency: 33.1808 ms

Category Match Rate: 0.80

Top Result: US no-show: Teamwork When the NBA began sending its players to the Olympics in 1992, one of the league #39;s goals was to spread basketball around the globe. Clearly, the idea worked.

